RESTAURANT BUDGET PLANNER

Java Swing and Oracle Database Project Report

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Abstract

This project, Restaurant Budget Planner, is a desktop application intended to help a restaurant organize and review its financial information. It separates planned spending from actual expenses and records revenue so users can review financial activity for a selected period.

The application is built with Java, Java Swing, JDBC, and Oracle Database. It has Manager and Accountant roles. The Manager prepares budgets and reviews financial information; the Accountant records day-to-day expenses and revenue. Totals and comparisons are calculated by the application from stored records rather than maintained as separate total records.

1. Introduction

Restaurants need to keep track of planned spending, actual expenses, and money earned. When these records are handled separately or manually, it can be difficult to compare a budget with actual activity and prepare a clear financial summary.

Restaurant Budget Planner provides a single desktop interface for recording and reviewing this information. The database stores individual records, while the application retrieves and calculates totals when needed.

2. Problem Statement

A restaurant needs a structured way to record budgets, expenses, and revenue and to compare planned spending with actual spending. The project addresses this need by providing a role-based application backed by an Oracle relational database.

3. Objectives

Provide authentication for a restaurant and its users.

Allow a Manager to create a budget for a selected period and enter multiple budget categories with planned amounts.

Allow an Accountant to record expenses and revenue.

Store records in Oracle Database through JDBC.

Calculate financial totals from saved records.

Support financial review and reporting using stored budget, expense, and revenue data.

4. Project Scope

The current application is a Java desktop application for one restaurant account at a time. It supports the Manager and Accountant roles. Restaurant registration creates the restaurant record and its associated user accounts. Restaurant-specific records are linked to a restaurant through the user and restaurant relationship.

The scope includes authentication, budget management, expense entry, revenue entry, financial calculations, and reporting. The application uses Oracle Database for persistent storage.

5. Technologies Used

- **Java:** application logic and service classes.
- **Java Swing:** desktop graphical user interface.
- **JDBC:** communication between Java and Oracle Database.
- **Oracle Database 21c XE:** relational data storage.
- **Oracle JDBC driver (ojdbc11.jar):** JDBC connectivity.
- **Visual Studio Code:** development environment.

6. User Roles

- **Manager:** creates budget plans, enters planned amounts by category, and reviews financial performance and reports.
- **Accountant:** records actual expenses and revenue and views transaction information and financial summaries.

The application verifies a user's role during authentication so the user is directed to the appropriate dashboard.

7. System Modules

- **Authentication Module:** supports restaurant registration and login, followed by Manager or Accountant login and role verification.
- **Budget Module:** allows the Manager to select a budget period, add categories and planned amounts, calculate the total planned budget, and save the budget.
- **Expense Module:** records actual expenses with information such as date, category, amount, description, and the user who recorded the transaction.
- **Revenue Module:** records revenue with its date, category, amount, description, and recording user.
- **Financial Analysis Module:** calculates totals and supports comparisons using stored budget, expense, and revenue records.
- **Report Module:** presents financial information for a selected period, including budget performance and financial summaries.

8. Authentication Workflow

Restaurant registration collects the restaurant name and password, the Manager username and password, and the Accountant username and password. The application stores the restaurant and user records in Oracle Database.

After restaurant login succeeds, the Welcome screen displays the restaurant name and allows the user to select Manager or Accountant. The selected role's login checks the username, password, and role before opening the appropriate dashboard.

The Java AuthenticationService handles authentication logic and uses the database connection layer to communicate with Oracle.

9. Budget Management Workflow

The Manager selects a budget period and enters one or more budget categories with planned amounts. The application validates the entries and calculates the total planned budget automatically.

The total is calculated from the detail amounts and is not stored as a separate total field. The application saves the overall budget in BUDGET and its category amounts in BUDGET_DETAILS. These related inserts are handled as a transaction so the budget is not left partially saved if an error occurs.

The Manager's user ID is obtained from the authenticated session rather than typed manually.

10. Expense and Revenue Management

The Accountant records actual expenses and revenue through the application. Each transaction includes a date, category, amount, description, and the user who recorded it. Expense records can also be associated with the applicable budget detail.

The application uses saved transaction records to display recent activity and calculate period totals. Queries are scoped to the authenticated restaurant to keep records separated between restaurant accounts.

11. Database Design

The database uses relational tables and primary-key/foreign-key relationships to connect users, budgets, budget details, expenses, and revenue. The multi-restaurant authentication design also includes RESTAURANT and APP_USER.

RESTAURANT stores restaurant login information. APP_USER stores Manager and Accountant accounts and links each account to a restaurant. BUDGET stores the overall budget plan and the Manager who created it. BUDGET_DETAILS stores category-level planned amounts. EXPENSE stores actual spending. REVENUE stores income records.

Primary keys uniquely identify records. Foreign keys preserve relationships between related records. Individual financial transactions are stored as records; totals are calculated when required.

12. Main Database Entities

- **RESTAURANT:** Stores restaurant identity and restaurant login information.
- **APP_USER:** Stores user login details, role, and restaurant association.
- **BUDGET:** Stores a budget plan and its period and creator.
- **BUDGET_DETAILS:** Stores the categories and planned amounts belonging to a budget.
- **EXPENSE:** Stores actual expenses and links the recording user and applicable budget detail.
- **REVENUE:** Stores revenue transactions and links the recording user.

13. Financial Calculations

Total expenses are calculated by summing expense amounts for the selected restaurant and period. Total revenue is calculated by summing revenue amounts for the same period.

Budget performance can be evaluated by comparing the planned amount for each category with the sum of related actual expenses. A remaining amount can be calculated as planned amount minus actual expenses. Net balance can be calculated as total revenue minus total expenses.

These values are derived from saved records. They do not need separate tables solely to store totals.

14. Application Structure

The project is organized into packages for the graphical interface, authentication, budget, expense, revenue, financial analysis, reports, and database connectivity. GUI classes collect user input and display results. Service classes handle application operations and database queries. DBConnection provides the JDBC connection to Oracle.

15. Validation and Transaction Handling

The application checks required fields and validates numeric amounts before saving. Budget amounts must be valid numeric values and cannot be negative. At least one budget category is required before a budget can be saved.

Saving a budget involves inserting a BUDGET record and related BUDGET_DETAILS records. A database transaction is used so all related inserts can be committed together; if an error occurs, the operation can be rolled back.

16. Testing

Project testing includes signing in with restaurant credentials, checking Manager and Accountant role verification, opening the relevant dashboards, entering budget categories and amounts, saving budgets, recording expense and revenue transactions, and reviewing the resulting database records.

The Budget Module was reported as tested with Oracle and displayed the message “Budget saved successfully!” The user has also reported connecting Accountant dashboard totals and recent transaction information to actual Oracle data. Final testing should confirm each workflow using the current project build and database.

17. Limitations and Future Improvements

The application is a desktop application and depends on a configured Oracle Database and JDBC driver. Final verification should be performed with the lecturer's expected environment and sample data.

Possible future improvements include stronger password handling, more detailed audit logging, additional automated tests, exportable reports, and further interface improvements. These are recommendations for future work, not claims that these features are already implemented.

18. Conclusion

Restaurant Budget Planner brings restaurant budget planning, expense recording, revenue recording, and financial review into one Java desktop application backed by Oracle Database. It separates Manager and Accountant responsibilities, stores individual financial records, and calculates totals from those records when needed.

The project demonstrates the use of Java Swing for the interface, JDBC for database access, and relational database relationships for organizing financial data.

Appendix A: Repository Checklist

- Java source folders and files
- Database SQL scripts
- Project report in DOCX and PDF formats
- README.md with project overview and run instructions
- Required JDBC driver or instructions for obtaining it, subject to lecturer and licensing requirements

Before submission, confirm that the report matches the final code and that no database passwords or other credentials are included in the repository.